

# Transportation Frequency and Employment Status

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The primary purpose of this manuscript is to learn and develop the skills and tools specific to Quarto documents. The simple 2x2 anova was used to examine the relationship between employment status, vehicle access, and total number of days per month an individual uses public transportation or rideshare.

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## Introduction:

The type of transportation an individual chooses to use is influenced by a variety of factors, including employment status and access to a vehicle. Workers in America typically drive a personal vehicle to work everyday (Yang et al. 2015). However, public transportation is also commonly used to commute to work, and rideshare services like Uber and Lyft have become increasingly popular in recent years.

Access to a vehicle greatly reduces the need for public transportation and rideshare services. I hypothesize that access to a vehicle will decrease the total number of public transportation and rideshare rides for both full-time employees and individuals with other employment statuses. The null hypothesis is that there is no difference in transportation usage between individuals with and without access to a vehicle.

A full-time employee that works from the office need to commute to the office 20-22 days per month, while part-time employees, contractors, and retirees will not need to commute as frequently. I hypothesize that full-time employed individuals will have higher transportation usage than those with other types of employment status whether or not they have access to a vehicle. The null hypothesis is that there is no difference in transportation usage between full-time employed individuals and those with other types of employment status.

## Data and Methods

The data was provided by professor Adam Nelson, PhD, at the University of Wisconsin, Madison for the students enrolled in the Data Science in Human Behavior masters program.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from IPython.display import display, Markdown
import statsmodels.api as sm
from statsmodels.formula.api import ols
from collections import defaultdict

sns.set_context('talk')

dataA = pd.read_csv('data_orig/PRCAProlificExport_FileA.csv')
dataB = pd.read_csv('data_orig/PRCAProlificExport_FileB.csv')
dataC = pd.read_csv('data_orig/PRCAQualtricsExport_FileC.csv')

questions = pd.read_csv('data_dictionary.csv')

# rename the columns to variable_name and question
questions.rename(columns={questions.columns[0]: 'variable_name',
↪ questions.columns[1]: 'question'}, inplace=True)

# merge dataframes
data_all = pd.merge(dataA, dataB, how = 'outer')
```

```
# Then merge the result with dataC (Q0 in dataC contains Participant id that
↪ matches Participant id in dataA/dataB)
data_all = pd.merge(data_all, dataC, left_on='Participant id', right_on='Q0',
↪ how='right')
```

Source: [Article Notebook](#)

## Data Cleaning

Source: [Article Notebook](#)

```
#create an indexing column because participant id a long string of numbers
↪ and we want to simplify it. This index will be used as the participant
↪ id. Then move it to the first column

data_all['p_id'] = data_all.index
data_all = data_all[['p_id'] + [col for col in data_all.columns if col !=
↪ 'p_id']]

#drop columns that are not needed
df = data_all.drop(columns=['EndDate', 'StartDate', 'LocationLatitude',
↪ 'LocationLongitude', 'Started at', 'Completed at', 'ResponseId',
↪ 'Participant id', 'Submission id', 'Total approvals', 'Country of
↪ residence', 'Student status'])

#filter for participants that took more than 30 seconds to complete the
↪ survey

df = df[df['Duration (in seconds)'] > 30]

df = df.drop(columns=['Duration (in seconds)', 'Time taken'])
```

Source: [Article Notebook](#)

```
# Get unique values for each Q column (excluding Q0)
q_unique_vals = {}
for col in df.columns:
    if col.startswith('Q') and not col.startswith('Q0'):
        unique_vals = tuple(sorted(df[col].dropna().unique()))
```

```

        q_unique_vals[col] = unique_vals

# Group columns by their unique values
value_groups = defaultdict(list)
for col, vals in q_unique_vals.items():
    value_groups[vals].append(col)

group2_mapping = {
    '0-1 days a month': 1,
    '2-4 days a month': 2,
    '4-8 days a month': 3,
    '8 or more days a month': 4,
    'Never': 0
}
group3_mapping = {
    '1-2 rides in a typical day': 1,
    '3-4 rides in a typical day': 2,
    '5-6 rides in a typical day': 3,
    '7 or more rides in a typical day': 4
}

for col in df.columns:
    if col in value_groups[tuple(sorted(group3_mapping.keys()))]:
        df[col + '_numeric'] = df[col].map(group3_mapping)

for col in df.columns:
    if col in value_groups[tuple(sorted(group2_mapping.keys()))]:
        df[col + '_numeric'] = df[col].map(group2_mapping)

#rename the new numeric columns something that makes more sense
df = df.rename(columns={
    'Q26_numeric': 'public_trans_per_month',
    'Q27_numeric': 'public_trans_reg_day',
    'Q28_numeric': 'rideshare_per_month',
    'Q29_numeric': 'rideshare_reg_day',
    'Q21': 'Car Access'})

#drop ALL Q columns
df = df.drop(columns=[col for col in df.columns if col.startswith('Q')])

```

```
#remove 1 participant that has 'Not Sure' as their car access response
df = df[df['Car Access'] != 'Not Sure']

# create a new column called Full-Time that combines Part-Time and Other into
↳ Other for Employment Status
df['Full-Time'] = df['Employment status'].apply(lambda x: 'Other' if x in
↳ ['Part-Time', 'Other'] else x)

# create my new outcome variable - total rides
df['total_rides'] = df['public_trans_per_month'] + df['rideshare_per_month']

df = df.drop(columns=['public_trans_reg_day', 'rideshare_reg_day'])
```

Source: [Article Notebook](#)

```
num_rows = df.shape[0]
```

Source: [Article Notebook](#)

Participant IDs from the original data set were long complicated strings of letters and numbers, so I created a new simplified participant ID as an index and removed the old long string column.

Following some EDA, I noticed 2 participants completed the survey in less than 30 seconds. That is an unreasonably short amount of time to answer all of the questions, so I removed those participants from the data set, as their responses will not be accurate.

Next I reviewed my independent variables **Car Access** and **Employment status**. To simplify my analysis, I converted **Car Access** into a binary variable (yes/no) and **Employment status** into a binary variable (full-time/other). There was one participant with “Not Sure” as their response for access to a vehicle, so they were removed from the data set. Participants that listed part-time or other employment status were combined into other employment status.

My dependent variable **total\_rides** was created by converting the **rideshare\_per\_month** and **public\_trans\_per\_month** variables from categorical to numeric and then summing the two variables.

Finally, I dropped the variables in the data set that are not used in this analysis. The final cleaned dataset contains 270 participant responses.

Model assumptions were checked in the [supplemental document](#).

Source: [Article Notebook](#)

```
# export the data frame to a csv so I can load it in supplemental.qmd
df.to_csv('cleaned_data.csv', index=False)
```

Source: [Article Notebook](#)

## Descriptive Statistics

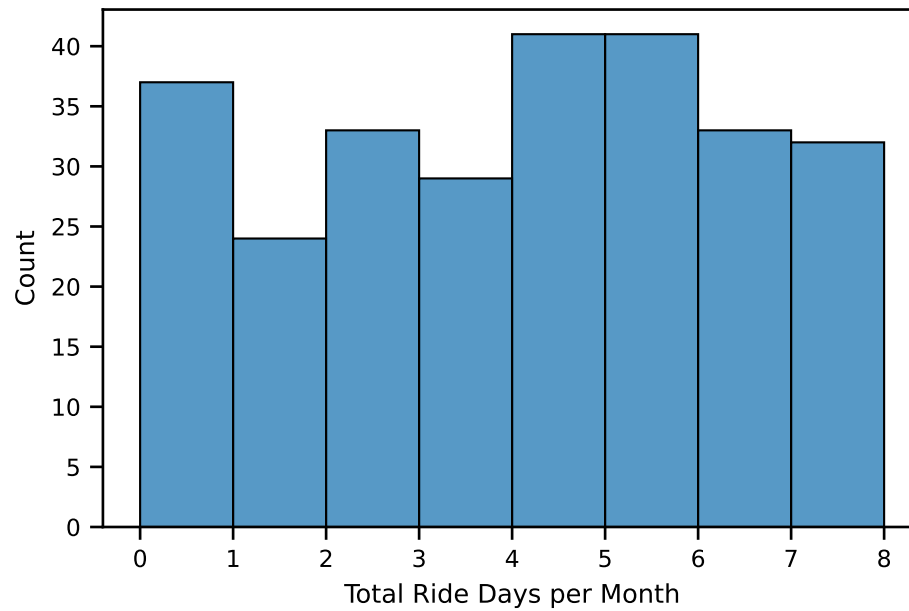
Source: [Article Notebook](#)

```
# Drop p_id column
# df_plot = df.drop(columns=['p_id'])

# df_plot.hist(figsize=(8, 10))
sns.set_context('paper')
sns.histplot(data=df, x='total_rides', bins=8)
plt.xlabel('Total Ride Days per Month')

display(plt.show())
```

Figure 1



(a) Total Ride Days

None

(b)

Source: [Article Notebook](#)

```
# Check for any unreasonable values in the dataset

# Create a mapping of variable names to human-readable descriptions
variable_descriptions = {
    'public_trans_per_month': 'Public Trans Use',
    'rideshare_per_month': 'Rideshare Use',
    'total_rides': 'Total Ride Use',
    'Age': 'Age'
}

# Get the descriptive statistics and transpose
desc_stats = df.drop(columns = ['p_id']).describe().transpose().apply(lambda
    ↪ x: round(x, 2))
```

```
# Add human-readable descriptions as a new column
desc_stats['Variable'] = desc_stats.index.map(variable_descriptions)

# Drop the existing index and set the new column as the index
desc_stats =
    ↪ desc_stats.reset_index(drop=True).set_index('Variable').drop(columns =
    ↪ ['25%', '50%', '75%'])

display(desc_stats)
```

Table 1: Descriptive statistics for the cleaned dataset

|                  | count | mean  | std   | min  | max  |
|------------------|-------|-------|-------|------|------|
| Variable         |       |       |       |      |      |
| Age              | 251.0 | 39.32 | 13.38 | 17.0 | 77.0 |
| Public Trans Use | 270.0 | 2.13  | 1.46  | 0.0  | 4.0  |
| Rideshare Use    | 270.0 | 1.51  | 1.24  | 0.0  | 4.0  |
| Total Ride Use   | 270.0 | 3.63  | 2.35  | 0.0  | 8.0  |

Source: [Article Notebook](#)

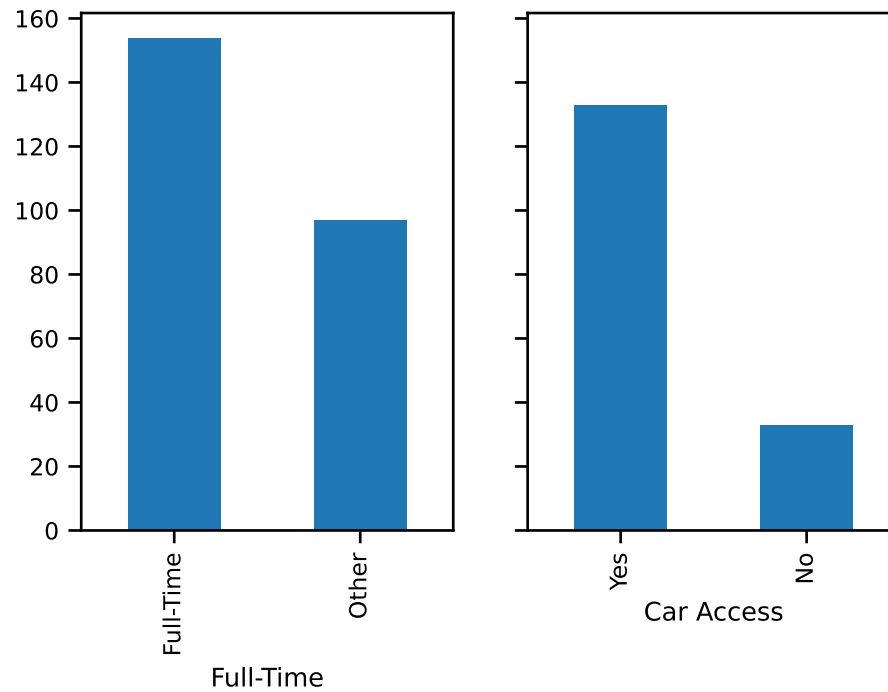
```
# put value count for employmentstats and for car access in the same figure
    ↪ and make the y axis the same for each plot and don't show the y axis for
    ↪ the 2nd plot
fig, axes = plt.subplots(1, 2, sharey=True)

df['Full-Time'].value_counts().plot(kind='bar', ax=axes[0], rot=90)

df['Car Access'].value_counts().reindex(['Yes', 'No']).plot(
    kind='bar',
    ax=axes[1],
    rot=90
)
plt.show()
```



Figure 2: Bar Plots of Employment Status and Vehicle Access



Source: [Article Notebook](#)

```
# -----  
# Employment status  
# -----  
  
employment_descriptions = {  
    'Full-Time': 'Full-Time Employment',  
    'Other': 'Other Employment Status'  
}  
  
employ_counts = df['Full-Time'].value_counts()  
  
employ_tbl = pd.DataFrame({'count': employ_counts})  
  
employ_tbl['Employment'] = employ_tbl.index.map(  
    employment_descriptions  
)
```

```

employ_tbl = (
    employ_tbl
    .reset_index(drop=True)
    .set_index('Employment')
)

employ_tbl.loc['Total'] = employ_tbl['count'].sum()

# -----
# Vehicle access
# -----

VehicleAccess = {
    'Yes': 'Has Access to Vehicle',
    'No': 'No Access to Vehicle'
}

car_counts = df['Car Access'].value_counts()

car_tbl = pd.DataFrame({'count': car_counts})

car_tbl['Vehicle Access'] = car_tbl.index.map(
    VehicleAccess
)

car_tbl = (
    car_tbl
    .reset_index(drop=True)
    .set_index('Vehicle Access')
)

car_tbl.loc['Total'] = car_tbl['count'].sum()

# -----
# Combine the two tables
# -----

employ_tbl = employ_tbl.reset_index()
car_tbl = car_tbl.reset_index()

# Make the tables the same number of rows

```

```

max_rows = max(len(employ_tbl), len(car_tbl))

employ_tbl = employ_tbl.reindex(range(max_rows))
car_tbl = car_tbl.reindex(range(max_rows))

# Rename columns
employ_tbl.columns = ['Employment Status', 'Count']
car_tbl.columns = ['Vehicle Access', 'Count']

# Combine side-by-side
combined_tbl = pd.concat(
    [employ_tbl, car_tbl],
    axis=1
)

combined_tbl = combined_tbl.reset_index(drop = True)
combined_tbl

```

Table 2: Value Counts of Employment Status and Car Access

|   | Employment Status       | Count | Vehicle Access        | Count |
|---|-------------------------|-------|-----------------------|-------|
| 0 | Full-Time Employment    | 154   | Has Access to Vehicle | 133   |
| 1 | Other Employment Status | 97    | No Access to Vehicle  | 33    |
| 2 | Total                   | 251   | Total                 | 166   |

Source: [Article Notebook](#)

```

grouped = df.groupby(['Full-Time', 'Car
↳ Access'])['total_rides'].describe().drop(columns = ['25%', '50%',
↳ '75%']).round(2)
grouped

```

Table 3: Descriptive Statistics for Total Rides by Employment Status and Vehicle Access

|           |            | count | mean | std  | min | max |
|-----------|------------|-------|------|------|-----|-----|
| Full-Time | Car Access |       |      |      |     |     |
| Full-Time | No         | 10.0  | 4.80 | 2.04 | 0.0 | 7.0 |
|           | Yes        | 75.0  | 3.49 | 2.16 | 0.0 | 8.0 |
| Other     | No         | 16.0  | 4.31 | 1.58 | 2.0 | 8.0 |

Table 3: Descriptive Statistics for Total Rides by Employment Status and Vehicle Access

|           |            | count | mean | std  | min | max |
|-----------|------------|-------|------|------|-----|-----|
| Full-Time | Car Access |       |      |      |     |     |
|           | Yes        | 51.0  | 2.14 | 2.14 | 0.0 | 8.0 |

Source: [Article Notebook](#)

Table 1 give general descriptive statistics of the numeric values of the data, with the mean Total Ride Days per Month equal to 3.63.

Figure 1 shows the distribution of total ride days per month. This is a uniform distribution and should be noted.

Figure 2 depicts the counts of participants by employment status and vehicle access and Table 2 shows the actual value counts. It should be noted that there are missing values in the data, specifically for the **Car Access** variable, so that is the reason for the different total values.

Table 3 shows the descriptive statistics of my dependent variable **total\_rides** by employment status and vehicle access.

## Methods

I conducted a 2x2 ANOVA to examine the effects of employment status and vehicle access on the total number of rides.

## Results

Source: [Article Notebook](#)

```
# Create the ANOVA model (using Q() for column names with special characters)
df['full_time'] = df['Full-Time']
df['car_access'] = df['Car Access']

model = ols('total_rides ~ C(full_time) + C(car_access)', data=df).fit()
```

Source: [Article Notebook](#)

```
# Get the ANOVA table
anova_table = sm.stats.anova_lm(model, typ=3).round(2)

anova_table.index = ['Intercept', 'Full-Time', 'Car Access', 'Residual']

# Display the ANOVA tables
anova_table
```

Table 4: 2x2 ANOVA results for total rides by employment status and vehicle access

|            | sum_sq | df    | F      | PR(>F) |
|------------|--------|-------|--------|--------|
| Intercept  | 563.22 | 1.0   | 128.38 | 0.0    |
| Full-Time  | 53.43  | 1.0   | 12.18  | 0.0    |
| Car Access | 68.83  | 1.0   | 15.69  | 0.0    |
| Residual   | 653.68 | 149.0 | NaN    | NaN    |

Source: [Article Notebook](#)

```
# Get the parameter estimates
params = model.params.round(2)
conf_int = model.conf_int().round(2)
pvalues = model.pvalues.round(3)

# Create a summary table
param_table = pd.DataFrame({
    'Estimate': params,
    '2.5%': conf_int[0],
    '97.5%': conf_int[1],
    'P-value': pvalues
})

#rename the indexes to better names
param_table.index = ['Intercept', 'Full-Time', 'Car Access']

param_table
```

Table 5: Parameter estimates for the ANOVA model

|            | Estimate | 2.5%  | 97.5% | P-value |
|------------|----------|-------|-------|---------|
| Intercept  | 5.24     | 4.33  | 6.16  | 0.000   |
| Full-Time  | -1.21    | -1.89 | -0.52 | 0.001   |
| Car Access | -1.81    | -2.71 | -0.91 | 0.000   |

Source: [Article Notebook](#)

```
#create a boxplot of total_rides by employment status and vehicle access
sns.set_context('paper')
sns.boxplot(data=df, x='Full-Time', y='total_rides', hue='Car Access')
plt.title('Total Rides by Employment Status and Vehicle Access')
plt.xlabel('Employment Status')
plt.ylabel('Total Ride Use')
plt.legend(title='Vehicle Access', bbox_to_anchor=(1.05, 1), loc='upper
↪ left')

plt.tight_layout()
plt.show()
```

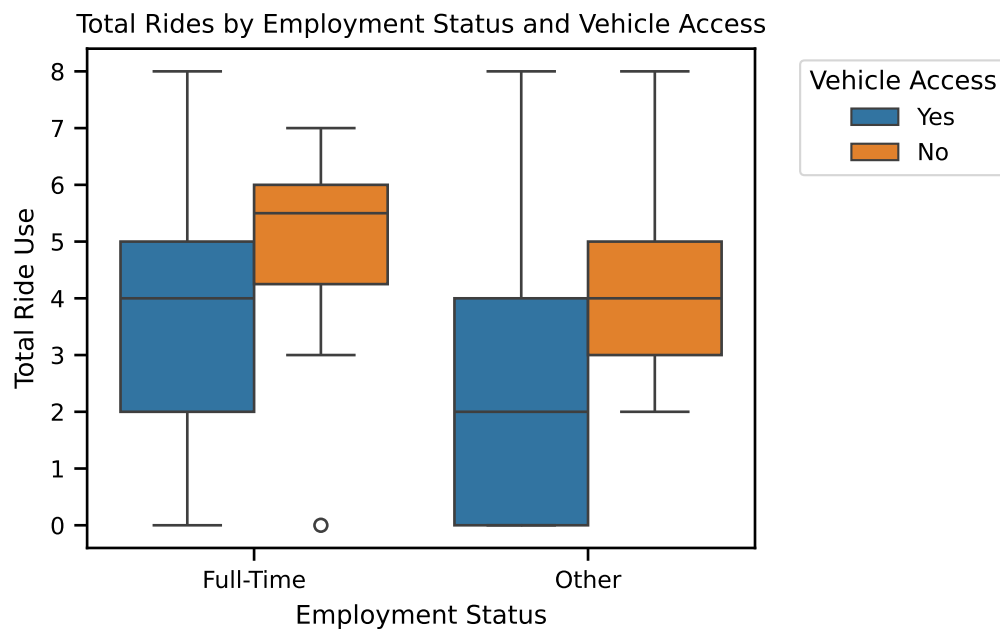


Figure 3: Boxplot of total rides by employment status and vehicle access

Source: [Article Notebook](#)

There is a significant main effect of **Full-Time** on **total\_rides** ( $F(1, 149) = 12.18, p < 0.001$ ), such that full-time employees have significantly more total ride days per month than other employees.

There is a significant main effect of **Car Access** on **total\_rides** ( $F(1, 149) = 15.69, p < 0.0$ ), such that individuals with car access have significantly less total ride days per month than other employees.

Results are also depicted visually in [Figure 3](#).

The parameter estimates referred to in [Table 5](#) are not interpretable due to converting total ride days per month from a categorical variable to a numerical variable. This is discussed in the limitations section ([Section 3](#)).

## Conclusion, Future Directions, & Limitations

Full Time employees have significantly more total ride days per month than other employees, and individuals with car access have significantly less total ride days per month than other employees([Figure 3](#)).

### Future Directions

Reviewing [Figure 3](#), there appears to be an outlier that is a full-time employee without access to a car with very low ride share and public transportation usage days. It would be interesting to further examine this outlier; however, it seems to be this person may walk to work or work from home.

Additional analysis should be done to examine if there is a significant effect of Full-time employment status on total ride days per month, such that the effect of car access on total rides per month is different for full-time employees versus part-time employees. This is a simple interaction effect that could be tested.

### Limitations

The distributions of the variables([Figure 2](#), [Figure 1](#)) are not normal, which may impact the results of the ANOVA test. Additionally, the data groups are unbalanced, as seen in [Table 3](#), which may also impact the results of the ANOVA test.

The values of **total\_rides** are calculated from categorical variables **ride\_share\_per\_month** and **public\_trans\_per\_month** that have unbalanced ranges: 0-1 days a month, 2-4 days a month, 4-8 days a month, 8 or more days a month, Never. Calculating **total\_rides** as the

sum of these ranges means that the value could represent a different range of actual ride counts. For example: if a participant has a total ride value of 4, they could have 4 ride share days and 0 public transportation days, or 3 ride share days and 1 public transportation day (visa versa), and 2 ride share days and 1 public transportation day. But, when calculating the actual number of days per month, the result could be 8+, 4-9, or 4-8. Therefore, the parameter estimates displayed in Table 5 are not informative values.

Source: [Article Notebook](#)

Yang, Lin, J. Aaron Hipp, Deepti Adlakha, Christine M. Marx, Rachel G. Tabak, and Ross C. Brownson. 2015. "Choice of Commuting Mode Among Employees: Do Home Neighborhood Environment, Worksite Neighborhood Environment, and Worksite Policy and Supports Matter?" *Journal of Transport & Health* 2 (2): 212–18. <https://doi.org/https://doi.org/10.1016/j.jth.2015.02.003>.